ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** independent variable	The quantity changed by the investigator is the independent variable.
2	**(a)** Robert Hooke	Hooke saw box-like compartments and called them cells.
3	**(b)** physical, mental and social well-being	Health is more than the absence of disease.
4	**(b)** cell or battery	A cell converts chemical energy to electrical energy.
5	**(d)** push or a pull	Forces are pushes or pulls.
6	**(a)** force acting per unit area	P = F / A.
7	**(a)** tiny particles	Matter consists of very small particles.
8	**(a)** element	That is the definition of an element.
9	**(b)** solution	Uniform composition.
10	**(c)** lie in the same plane	All three are coplanar.
11	**(b)** rotation of the Earth on its axis	Earth spins from west to east.
12	**(d)** living and non-living components interacting with each other	Biotic and abiotic parts together.
13	**(c)** water, air and a suitable temperature	Conditions needed for life are present.
14	**(c)** a question or curiosity about an observation	Questions drive investigations.
15	**(a)** cell, tissue, organ, organ system, organism	Cells form tissues; tissues form organs, and so on.
16	**(c)** something a doctor can observe or measure, such as a rash	Signs are observable.
17	**(d)** nichrome	Nichrome has high resistance and a high melting point.
18	**(d)** rough surfaces	Rough surfaces interlock.
19	**(b)** barometer	Barometer.
20	**(c)** melting	Fusion.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Testable explanation or guess before an experiment (1); e.g. a puri puffs more in hotter oil (1).	1+1
22	Robert Hooke in 1665 (1); He observed thin slices of cork under a microscope (1).	1+1
23	State of complete physical, mental and social well-being (1); Not merely the absence of disease (1).	1+1
24	A closed path along which electric current flows (1); Includes a source, wires and a device (1).	1+1
25	A push or a pull (1); Newton (N) (1).	1+1
26	Force per unit area; P = F/A (1); SI unit pascal (Pa) (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Question and hypothesis: hotter oil makes puffing faster (1); Vary temperature; keep dough, thickness and size same (1); Measure time with stopwatch, repeat, record in a table (1).	1+1+1
28	Peel a thin layer and place it in water on a slide (1); Add a drop of stain and cover with a cover slip carefully (1); Observe under the microscope and draw the rectangular cells (1).	1+1+1
29	Physical: body works well, e.g. energy for play (1); Mental: emotional balance, e.g. managing stress (1); Social: good relationships, e.g. helping neighbours (1).	1+1+1
30	Cell, switch and bulb connected in a loop with wires (1); When the switch is closed the path is complete (1); Current flows through the filament and the bulb glows (1).	1+1+1
31	Force can squeeze, stretch or bend (1); Pressing dough, stretching a rubber band (1); Shape change depends on force and material (1).	1+1+1

	OR: P = F / A (1); = 600 / 0.5 (1); = 1200 Pa (1).	1+1+1
32	Dissolve a spoon of salt/sugar in a beaker of water (1); Level of water hardly rises (1); Solute particles fit in the gaps between water particles (1).	1+1+1
	OR: Lustre: gold (1); Malleability: aluminium foil (1); Conductivity: copper wire (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Question stated clearly (1); Hypothesis and independent/dependent variables (1); Constants: oil temperature, thickness, size, frying time (1); Procedure: vary water, fry, observe puffing/time (1); Repeat and record in a table (1).	1+1+1+1+1
34	Cell wall: shape and protection (1); Cell membrane: controls entry/exit (1); Cytoplasm: site of cell activities (1); Nucleus: controls cell and has genetic material (1); Chloroplast and large vacuole: food making; storage (1).	1+1+1+1+1
35	WHO: complete well-being, not just absence of disease (1); Physical: normal body functions (1); Mental: positive emotions, coping with stress (1); Social: healthy relationships (1); All aspects are linked (1).	1+1+1+1+1
36	Set-up: wire, cell, switch, compass (1); Needle deflects when current flows (1); Conclusion: current has a magnetic effect (1); Reverse current: deflection reverses (1); Needle returns when current is off (1).	1+1+1+1+1
	OR: Moves a stationary body: kicking a ball (1); Stops a moving body: catching a ball (1); Changes direction: hockey stick hits (1); Changes speed: pedalling faster (1); Changes shape: squeezing clay (1).	1+1+1+1+1
37	P = F/A (1); Larger force or smaller area gives greater pressure (1); 50/0.25 = 200 Pa (1); 50/0.5 = 100 Pa (1); Examples: knife, camel's feet (1).	1+1+1+1+1
	OR: Dissolving sugar (1); Spreading of ink (1); Smell of perfume (1); Spaces between particles (1); Particles in constant motion (1).	1+1+1+1+1
38	Composition (1); Separation method (1); Properties (1); Examples (1); Fixed ratio or not (1).	1+1+1+1+1
	OR: Particle size (1); Appearance (1); Settling (1); Filtration (1); Examples (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER

ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** malleability	Malleable.
2	**(d)** brass	Alloy.
3	**(d)** diffuse reflection	Irregular.
4	**(a)** 27 days	Orbital period of the Moon.
5	**(d)** food chain	Linear sequence.
6	**(a)** nitrogen	About 78 per cent.
7	**(d)** length of the pendulum	Length is the quantity that is changed.
8	**(a)** vacuole	A large vacuole stores water and substances.
9	**(b)** vitamin C	Bleeding gums and weak joints.
10	**(c)** a resistor	Resistor symbol.
11	**(c)** positive and negative	Two kinds of charge.
12	**(b)** very low pressure	Low pressure centre.
13	**(a)** increase in temperature	Warmer liquids evaporate faster.
14	**(c)** a combination of two or more substances not chemically combined	Components keep properties.
15	**(b)** (mass of solute / mass of solution) x 100	Standard definition.
16	**(c)** R = 2f	Focus is midway.
17	**(d)** solar calendar	Gregorian.
18	**(b)** decomposition	Recycling.
19	**(c)** a small part of the total water	About 3 per cent is fresh and little is accessible.
20	**(d)** dependent variable	It is the quantity that is measured as the result.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Matter is made of tiny particles (1); Particles move and have spaces between them (1).	1+1
22	Pure substance made of only one kind of atom (1); e.g. iron, oxygen (1).	1+1
23	Solute is dissolved, solvent dissolves it (1); Solution is their homogeneous mixture (1).	1+1
24	Incident ray, reflected ray and normal lie in the same plane (1); Angle of incidence equals angle of reflection (1).	1+1
25	Earth rotates from west to east (1); So the Sun appears to move east to west (1).	1+1
26	Community of living organisms interacting with their physical environment (1); Pond, forest (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Add salt/sugar in portions to 100 mL of cold and of warm water (1); Count spoons dissolved till saturation (1); More dissolves in warm water (1).	1+1+1
28	Incident ray strikes mirror at the point O (1); Normal drawn at O (1); Reflected ray makes angle r = i on the other side (1).	1+1+1
29	Fix a vertical stick on level ground in the Sun (1); Mark the end of its shadow every hour (1); Shadow positions show the time of the day (1).	1+1+1
30	Biotic: algae, fish, frogs, bacteria (1); Abiotic: water, light, temperature, minerals (1); They interact with each other (1).	1+1+1
31	Crust: thin and solid (1); Mantle: thick, hot, semi-molten rock (1); Core: outer liquid, inner solid, iron and nickel (1).	1+1+1
	OR: Real work often moves back and forth (1); A surprising result may send scientists back to change the question (1); So the steps are repeated and revised (1).	1+1+1
32	All living bodies are made of cells (1); Each cell performs life functions (1); New cells arise from existing cells (1).	1+1+1

	OR: Vitamin A: night blindness (1); Vitamin C: scurvy, bleeding gums (1); Iron: anaemia, weakness (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Incident and reflected ray on same side of normal (1); i = r (1); Coplanar (1); Activity: pins and protractor on paper (1); Measured angles are equal (1).	1+1+1+1+1
34	Rotation: day and night (1); Revolution: year (1); Axis tilt 23.5 degrees (1); Seasons (1); Day length changes (1).	1+1+1+1+1
35	Biotic: producers, consumers, decomposers (1); Abiotic: light, water, air, soil (1); Producers need abiotic factors (1); Consumers depend on producers (1); Decomposers recycle matter (1).	1+1+1+1+1
36	Crust: solid, thin (1); Mantle: thick, hot rock (1); Outer core: liquid iron and nickel (1); Inner core: solid (1); Studied through earthquake waves (1).	1+1+1+1+1
	OR: Parallax or wrong reading of scale (1); Faulty or uncalibrated instrument (1); Repeat the readings (1); Take the average (1); Use proper instrument and procedure (1).	1+1+1+1+1
37	Bacteria: single-celled, no nucleus (1); Viruses: non-cellular, parasites (1); Fungi: lack chlorophyll; yeast, mushroom (1); Protozoa: single-celled, with nucleus; Amoeba (1); Each can cause disease in some cases (1).	1+1+1+1+1
	OR: Personal: handwashing, bathing, clean clothes (1); Community: clean water (1); Community: waste disposal (1); Community: drainage and sanitation (1); Together they break the chain of infection (1).	1+1+1+1+1
38	Crane: current ON lifts scrap (1); Current OFF drops it (1); Magnet separates iron from rubbish (1); Soft iron is easily magnetised and demagnetised (1); Steel would stay magnetised (1).	1+1+1+1+1
	OR: Leaves hang together when uncharged (1); Touching with a charged rod gives charge to all parts (1); Leaves carry the same charge and repel (1); More charge, more divergence (1); Used to detect charge (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER

ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** microscope	A microscope magnifies tiny things.
2	**(b)** getting along well with family and neighbours	Social health concerns relationships.
3	**(b)** a gap in the circuit	An open switch breaks the path.
4	**(c)** newton	Named after Isaac Newton.
5	**(d)** pascal	1 Pa = 1 N/m squared.
6	**(d)** sugar dissolves in water without raising the water level much	Sugar particles occupy spaces between water particles.
7	**(c)** O_2	Oxygen contains atoms of only one kind.
8	**(d)** solute	Substance dissolved.
9	**(b)** angle of incidence equals angle of reflection	i = r.
10	**(a)** 24 hours	One day.
11	**(a)** sunlight	Non-living.
12	**(b)** crust	Thin solid layer.
13	**(c)** a flexible approach that may be repeated and revised	Real investigations loop back and revise earlier steps.
14	**(d)** tissue	A tissue is a group of similar cells.
15	**(b)** chronic	Chronic diseases are long lasting.
16	**(b)** low melting point	It melts and breaks the circuit during overload.
17	**(c)** lubricants	Oil forms a layer.
18	**(c)** wind	Wind is air in motion.
19	**(d)** evaporation	Surface process.
20	**(a)** ductility	Ductile.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Crust (1); Mantle and core (1).	1+1
22	To check that the result is not a chance event (1); To get a more dependable (average) value (1).	1+1
23	It contains the genetic material (DNA) (1); It controls all activities of the cell (1).	1+1
24	They prepare the body to fight a particular pathogen (1); They prevent the disease before it occurs (1).	1+1
25	It has a very high melting point (1); It can glow without melting (1).	1+1
26	To reduce friction (1); So the coins slide easily (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	It is connected in series (1); Excess current heats and melts it (1); The circuit breaks and the appliances are safe (1).	1+1+1
28	A metal rod with two thin leaves in a jar (1); Touching a charged body spreads charge to leaves (1); Like charges repel; leaves diverge (1).	1+1+1
29	More air particles collide with the walls (1); Gas pressure increases inside (1); Balloon stretches (1).	1+1+1
30	Fastest particles escape from the liquid (1); Remaining particles have lower energy (1); Temperature of the liquid falls (1).	1+1+1
31	H and O are chemically combined (1); In a fixed ratio by mass (1); Water has properties different from H2 and O2 (1).	1+1+1
	OR: Mass of solute = 10/100 x 200 (1); = 20 g (1); Mass of water = 180 g (1).	1+1+1
32	Reflecting surface curves inwards vs outwards (1); Concave converges rays; convex diverges (1); Concave can form real images; convex only virtual (1).	1+1+1

	OR: Earth's axis is tilted by 23.5 degrees (1); The tilt keeps the same direction as Earth revolves (1); Sun's rays fall differently, changing day length (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Measures air pressure (1); Falls with altitude (1); Low pressure signals storms (1); High pressure signals clear weather (1); Used in weather forecast (1).	1+1+1+1+1
34	Wrap cotton on the bulb (1); Dip in water, then in acetone (1); Note the temperature fall (1); Acetone gives more cooling (1); Faster evaporation leads to more cooling (1).	1+1+1+1+1
35	H2: 2 hydrogen atoms (1); O2: 2 oxygen atoms (1); H2O: 2 H + 1 O (1); CO2: 1 C + 2 O (1); Molecule of an element vs compound (1).	1+1+1+1+1
36	Solid in liquid: sugar in water (1); Gas in liquid: soda water (1); Liquid in liquid: alcohol in water (1); Gas in gas: air (1); Solid in solid: brass (1).	1+1+1+1+1
	OR: Convex converges (1); Concave diverges (1); Focus and focal length (1); Convex: magnifier, camera (1); Concave: spectacles for short sight (1).	1+1+1+1+1
37	Sunlit half always faces the Sun (1); Moon orbits the Earth (1); Angle between Sun and Moon changes (1); We see varying fraction of the lit half (1); Full moon when Earth is between (1).	1+1+1+1+1
	OR: Food web is disturbed (1); Prey populations rise (1); Vultures: carcasses are not cleaned (1); Disease spreads (1); Balance is lost (1).	1+1+1+1+1
38	Absorbs UV (1); CFCs destroy ozone (1); Ozone hole over Antarctica (1); Montreal Protocol banned CFCs (1); Recovery is underway (1).	1+1+1+1+1
	OR: Observe and ask a question (1); Form a hypothesis (1); Plan and perform a fair test (1); Analyse data and conclude (1); New questions arise, so steps repeat (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER